Clean Fixed Circles Obstruct Isolated-Orbit Determinants on a Heisenberg Nilmanifold

Route-A structural certificate C146

Abstract

We freeze an explicit integer-lattice automorphism of the three-dimensional Heisenberg group. Its horizontal action is hyperbolic, while its center is fixed pointwise. Every iterate therefore owns a clean central fixed circle, has singular ordinary isolated-orbit stability factor, and has zero Lefschetz number. The horizontal torus provides an exact isolated-point control through iterate twenty. These are source-side obstruction results, not a target or arithmetic construction.

1 The frozen lattice automorphism

Write the real Heisenberg group as $H = \mathbb{R}^3$ with

$$(x, y, z)(X, Y, Z) = (x + X, y + Y, z + Z + xY), \qquad \Gamma = \mathbb{Z}^3.$$

On the compact left quotient $N = \Gamma\backslash H$, freeze

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad q(x, y) = x(x - 1) + xy + \frac{y(y - 1)}{2},$$

and

$$\Phi(x, y, z) = (2x + y, x + y, z + q(x, y)). \tag{1}$$

The central correction is forced by the coordinate cocycle. Direct expansion gives

$$q(v + w) - q(v) - q(w) = 2xX + xY + Xy + yY,$$

which equals $(2x + y)(X + Y) - xY$. Thus (1) is a group homomorphism. Moreover, $q(m, n)$ is integral for integer m, n, so the map preserves Γ and descends to N.

2 Clean fixation and singular stability

The embedded central circle

$$C = \{[0, 0, z] : z \in \mathbb{R}/\mathbb{Z}\}$$

is fixed pointwise by Φ, hence by every positive iterate. The horizontal eigenvalues are $(3 \pm \sqrt{5})/2$, neither of whose positive powers is one. Consequently the horizontal fixed classes at each iterate are discrete, and C is a one-dimensional clean fixed component.

The derivative is block lower triangular. For every $n \geq 1$, its diagonal blocks are A^n and 1, so

$$\det(I - D\Phi^n) = \det(I - A^n)(1 - 1) = 0. \tag{2}$$

Thus the usual stability denominator for an isolated periodic orbit is singular at every period. We do not introduce a clean-family regularization.

Let $\alpha = dx$, $\beta = dy$, and $\gamma = dz - x\,dy$, with $d\gamma = -\alpha \wedge \beta$. Heisenberg cohomology has dimensions $(1, 2, 2, 1)$. Pullback has trace $\operatorname{tr}(A^n)$ in both middle degrees and acts as the identity in degrees zero and three. Therefore

$$L(\Phi^n) = 1 - \operatorname{tr}(A^n) + \operatorname{tr}(A^n) - 1 = 0. \tag{3}$$

3 Horizontal control and exact validation

On the horizontal torus, fixed classes form $\ker(A^n - I : \mathbb{T}^2 \to \mathbb{T}^2)$, so their number is

$$|\det(A^n - I)| = \operatorname{tr}(A^n) - 2 = L_{2n} - 2, \tag{4}$$

where L_k is the Lucas sequence. Here $A = Q^2$ for $Q = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, proving the last equality. Exact evidence records (4), (2), and (3) through $n = 20$.

The independent checker passes 683 assertions, the SymPy reconstruction passes 87 checks, byte replay passes, and all 29 hostile receipts (28 repaired- hash semantic mutations and one stale hash) are rejected.

4 Route-A boundary

The strict tuple is (A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT), overall ROUTE_A_EXPLORATORY. We claim no isolated primitive-orbit ledger, target divisor, target functional equation or counting law, arithmetic/local factor, Euler factor, root number, automorphy, natural quantum operator, Hilbert–Pólya construction, or Route-B authorization. The scope literal is NO_BAD_EULER_OR_ROOT_NUMBER.